Mini Project Report On

Machine Learning Based Framework for Predicting Urban Air Quality

Submitted in partial fulfillment of the requirements for the award of the degree of

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in

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CERTIFICATE

This is to certify that the mini project report entitled ***"Machine Learning Based Framework for Predicting Urban Air Quality"*** *is a bonafide record of the work done by* ***ANNA BASTIN (U2303046)***, ***BENISON ELDHO (U2303070)***, ***ARUN KOLLAPPALLIL BOBY (U2303056)***, ***ARPIT MATHEW ABRAHAM (U2303055)***, *submitted to the APJ Abdul Kalam Technological University in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology (B. Tech.) in Computer Science and Engineering during the academic year 2025-2026.*

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Abstract

Air pollution has become a serious environmental and public health concern, especially in rapidly growing urban areas. Monitoring pollution levels is important for ensuring environmental safety and public awareness. Traditional air quality monitoring systems depend heavily on expensive sensor networks and monitoring stations, which limits their widespread deployment. This project presents a simplified air quality prediction approach that utilizes computer vision and image processing techniques to analyze environmental conditions using camera input.

The proposed system captures real-time images through a camera and processes them using Python-based tools. Through image processing, the system extracts visual features such as haze intensity and reduced visibility from the captured frames. These features are then analyzed using machine learning algorithms to estimate the level of air pollution. The system operates as a simulation model that demonstrates how visual data can be interpreted to predict environmental conditions without relying completely on traditional sensor-based systems.

Compared to existing monitoring approaches, the proposed solution offers a more accessible and cost-effective method for environmental monitoring, especially in situations where installing dedicated hardware sensors may not be feasible. By integrating air quality prediction, computer vision, machine learning, and image processing, the project highlights the potential of intelligent vision-based systems in supporting modern environmental monitoring solutions.

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List of Abbreviations

AQI	Air Quality Index
PM2.5	Particulate Matter (2.5 micrometers)
CO	Carbon Monoxide
NH3	Ammonia
ML	Machine Learning
GBM	Gradient Boosting Machine
UI	User Interface
CSV	Comma Separated Values
IoT	Internet of Things

Chapter 1

Introduction

1.1 Background

Air pollution has become one of the most serious environmental problems affecting human health and ecological balance across the world. Rapid urbanization, increasing industrial activities, vehicular emissions, and construction work have significantly contributed to the rise in air pollutants. Harmful pollutants such as particulate matter (PM2.5), carbon monoxide (CO), and ammonia (NH3) can negatively impact human health when present in high concentrations. Continuous exposure to polluted air can lead to respiratory illnesses, cardiovascular diseases, reduced lung function, and other health complications. Therefore, monitoring air quality and understanding pollution levels has become increasingly important for protecting public health and maintaining environmental sustainability.

The Air Quality Index (AQI) is a widely used metric that represents the overall quality of air in a specific area. It converts complex air pollution data into a single value that indicates whether the air is safe or harmful for people. Traditionally, air quality monitoring is carried out using specialized monitoring stations equipped with advanced sensors that measure pollutant concentrations. However, such systems are often expensive to deploy and maintain, and they may not always provide flexible or accessible monitoring solutions for smaller-scale applications or educational projects.

With the advancement of data science and machine learning technologies, it has become possible to design intelligent systems that can analyze environmental parameters and estimate air quality more efficiently. Machine learning models can identify patterns in environmental data and predict AQI values based on pollutant levels. These models enable faster analysis and can support real-time monitoring systems.

The proposed Air Quality Prediction System aims to demonstrate a smart approach for estimating air quality using pollution parameters and machine learning techniques.

The system considers environmental variables such as PM2.5, CO, and NH3 as input parameters for predicting the Air Quality Index. In addition, the system includes a camera module that captures environmental images, which can help in visually observing atmospheric conditions.

To facilitate testing and demonstration, the system includes a simulation module that generates pollution data similar to real sensor readings. This allows the system to function even in the absence of physical air quality sensors. The simulated data is processed by a trained machine learning model that predicts the AQI value based on the provided parameters. The predicted air quality information is then displayed through a dashboard interface, allowing users to observe and analyze pollution levels in a simple and interactive manner.

Overall, the system demonstrates how machine learning and simulation techniques can be used to develop a basic air quality prediction framework. Such systems can help in understanding pollution trends and serve as a foundation for more advanced environmental monitoring solutions in the future.

Table 1.1: Air Quality Index (AQI) Classification Levels

AQI Range	**Category**	**Health Impact**
0–50	Good	Air quality is satisfactory and poses little or no risk.
51–100	Satisfactory	Air quality is acceptable but minor health concerns may exist for sensitive individuals.
101–200	Moderate	Sensitive groups may experience health effects while the general public is less likely to be affected.
201–300	Poor	Everyone may begin to experience health effects.
301–400	Very Poor	Health alert: everyone may experience more serious health effects.
401–500	Hazardous	Health warnings of emergency conditions affecting the entire population.

1.2 Problem Definition

The major problem addressed in this project is the lack of simple, affordable, and user-friendly systems for monitoring and predicting air pollution levels in real time. Existing air quality monitoring solutions are often complex, expensive, or not easily accessible to the

general public, making it difficult for individuals and small organizations to understand environmental conditions. As a result, people are unable to make informed decisions regarding health and safety, especially in areas where pollution levels fluctuate frequently.

To solve this problem, this project proposes the development of an Air Quality Prediction System that estimates the Air Quality Index (AQI) using key environmental parameters such as PM2.5, CO, and NH3. By leveraging a machine learning model, the system can analyze sensor data and provide accurate, real-time predictions of air quality. This approach aims to create a cost-effective, efficient, and accessible solution that helps users monitor pollution levels and take necessary precautions.

1.3 Scope and Motivation

1.4 Scope

The scope of this project is to develop an Air Quality Prediction System that estimates the Air Quality Index (AQI) using environmental parameters such as PM2.5, CO, and NH3. The system uses a machine learning model to analyze these parameters and generate AQI predictions. A simulation module is included to produce pollution data when real sensors are not available, allowing the system to be tested under different conditions. The project also includes a camera input that can capture environmental images for visual observation of atmospheric conditions. The predicted AQI values are displayed through a user interface, enabling users to easily view and analyze air quality levels.

1.5 Motivation

The motivation for this project arises from the growing problem of air pollution and its impact on human health and the environment. Poor air quality can cause serious respiratory and cardiovascular diseases, making it important to monitor pollution levels regularly. However, traditional air quality monitoring systems can be costly and not easily accessible in many areas. This project explores the use of machine learning and simulation techniques to create a simple and efficient air quality prediction system. By developing this system, the project aims to demonstrate how technology can help in understanding and monitoring environmental conditions more effectively.

1.6 Objectives

- To develop an Air Quality Prediction System that can estimate the Air Quality Index (AQI) based on environmental parameters such as PM2.5, CO, and NH3.
- To implement a machine learning model that analyzes pollution data and predicts AQI values accurately.
- To design a simulation module that generates pollution data in the absence of real air quality sensors.
- To integrate a camera input module for capturing environmental images to observe atmospheric conditions.
- To create a user-friendly interface that displays the predicted AQI values for easy monitoring and analysis.
- To demonstrate how modern technologies such as machine learning and data visualization can be applied to environmental monitoring systems.

1.7 Challenges

One of the main challenges in this project was obtaining reliable environmental data for training and testing the machine learning model. Another challenge was designing a simulation module that could generate realistic pollution values in the absence of real sensors. Integrating different components such as the machine learning model, camera input, and user interface into a single system also required careful design and implementation.

1.8 Assumptions

- It is assumed that the simulated pollution data used in the system represents realistic environmental conditions.
- The machine learning model is assumed to provide reasonably accurate AQI predictions based on the given input parameters such as PM2.5, CO, and NH3.
- It is assumed that the camera input can provide a basic visual representation of the surrounding environmental conditions.

1.9 Societal / Industrial Relevance

The Air Quality Prediction System has significant relevance for both society and industry as it helps in monitoring and understanding environmental pollution levels. For society, the system can be used to increase awareness about air quality conditions and the potential health risks associated with high pollution levels. By providing AQI predictions, individuals can take necessary precautions such as limiting outdoor activities during periods of poor air quality. The system can also be useful in educational institutions and research environments for studying environmental monitoring and data analysis.

From an industrial perspective, the system can assist industries and organizations in monitoring pollution levels around their facilities. Industries can use such systems to evaluate environmental impact and ensure that their operations remain within acceptable pollution limits. Additionally, the integration of machine learning and data visualization demonstrates how modern technologies can be applied to environmental monitoring solutions. Such systems can serve as a foundation for developing more advanced smart city or IoT-based air quality monitoring platforms in the future.

1.10 Organization of the Report

This report is organized into several sections that describe the development and implementation of the Air Quality Prediction System. The first chapter introduces the background, scope, motivation, objectives, challenges, and assumptions of the project. The second chapter presents the literature review and discusses related work in the field of air quality monitoring and prediction. The third chapter explains the system design and architecture, including the overall structure of the proposed system. The fourth chapter describes the implementation details, including the simulation module, machine learning model, and user interface. Finally, the last chapter presents the results, conclusions, and possible future improvements of the project.

Chapter 2

Software Requirements Specification

2.1 Introduction

The Software Requirements Specification (SRS) document presents a comprehensive description of the Air Quality Prediction System, including its objectives, scope, and overall functionality. It serves as a formal reference for developers, designers, and stakeholders to understand the system requirements and ensure that the final product meets the intended goals.

Air pollution has become a major concern in urban and semi-urban areas, affecting human health and environmental sustainability. However, many existing monitoring systems are either expensive or not easily accessible to the general public. This project aims to address this issue by developing a cost-effective and user-friendly system capable of predicting the Air Quality Index (AQI) using key environmental parameters such as PM2.5, CO, and NH3.

The system integrates sensor data acquisition, data preprocessing, and a machine learning model to analyze environmental conditions and generate accurate AQI predictions. It is designed to provide real-time insights that can help users make informed decisions regarding their health and surroundings. Additionally, the system may include visualization components and camera input to enhance the understanding of environmental conditions.

This document outlines both functional requirements, such as data collection, AQI prediction, and result display, and non-functional requirements, including performance, reliability, and usability. It also highlights system constraints, assumptions, and dependencies that influence the design and implementation.

The following figure illustrates the overall architecture of the system, showing how different modules interact with each other to achieve the desired functionality.

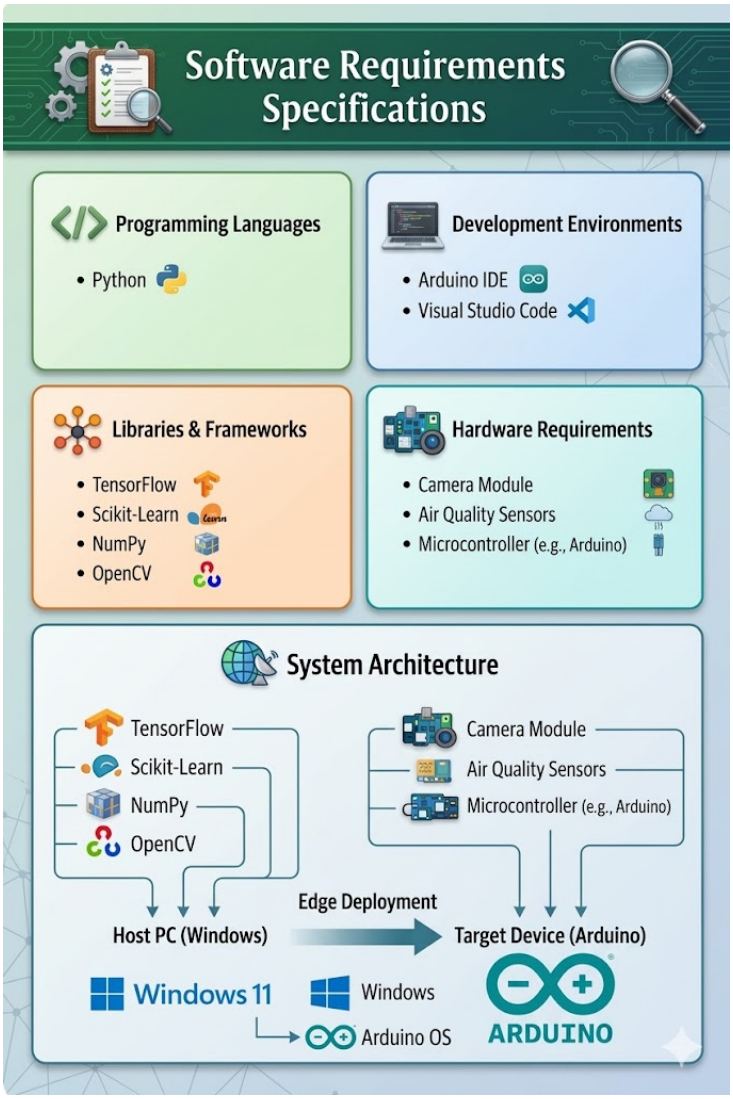


Figure 2.1: Software Requirements Specifications

2.2 Overall Description

The Air Quality Prediction System is designed to monitor and estimate air quality levels using environmental data and machine learning techniques. The system collects input data such as pollutant values and camera-based environmental observations, processes the data, and predicts the Air Quality Index (AQI). The predicted AQI helps in understanding the pollution level and its possible impact on human health. The system also provides a user interface where users can simulate pollution values and observe the predicted air quality results.

The system mainly consists of a simulation module, a camera input module, a machine learning prediction model, and a visualization interface. The simulation module allows

users to manually enter pollutant values to observe how different pollution levels affect the AQI. The camera module captures environmental images and extracts relevant information that contributes to air quality estimation. The machine learning model processes these inputs and predicts the AQI value. The results are displayed through an interactive user interface that presents the predicted air quality in an understandable format.

The proposed system aims to provide a simplified and educational platform for understanding air pollution monitoring and prediction. It demonstrates how environmental data, computer vision, and machine learning techniques can be integrated to build an intelligent environmental monitoring solution. The system can be further extended in the future to include real-time sensors, advanced prediction models, and integration with smart city monitoring platforms.

2.3 External Interface Requirements

2.3.1 User Interface

The system provides a graphical user interface that allows users to interact with the Air Quality Prediction System easily. The interface displays the predicted Air Quality Index (AQI) along with the corresponding air quality category. Users can input pollution values manually through the simulation module to observe how different pollutant levels affect the AQI. The interface also supports camera input to capture environmental conditions for prediction. The results are displayed through charts, numerical AQI values, and visual indicators to improve readability and user understanding.

2.3.2 Hardware Interface

The system interacts with external hardware components to collect environmental data. A camera module is used to capture environmental images that help estimate air pollution conditions. The system is also designed to support integration with air quality sensors such as gas sensors and particulate matter sensors in future extensions. These hardware devices provide input data that can be processed by the system to generate AQI predictions.

2.3.3 Software Interface

The system is implemented using Python and integrates several software libraries and frameworks. Libraries such as NumPy and Scikit-Learn are used for data processing and machine learning prediction. OpenCV is used for camera input and image processing. The system runs in a development environment such as Visual Studio Code and is compatible with operating systems including Windows and Linux. The user interface communicates with the prediction model to display AQI results based on the provided inputs.

You can insert equations into your file using the following code:

$$AQI = f(PM_{2.5}, CO, NH_3) \tag{2.1}$$

2.4 System Features

The Air Quality Prediction System includes several features that help in monitoring and estimating air quality levels using environmental data and machine learning techniques.

1. Air Quality Prediction: The system predicts the Air Quality Index (AQI) based on environmental parameters such as PM2.5, CO, and NH3. A trained machine learning model processes the input data and generates the predicted AQI value.

2. Simulation Module: The system includes a simulation feature that allows users to manually enter pollution values. This helps in testing the system and observing how different pollutant levels affect the predicted AQI.

3. Camera Input Module: The system uses a camera to capture environmental images. These images can help in observing atmospheric conditions and supporting the air quality analysis process.

4. Data Processing and Analysis: The system processes the input pollution parameters and prepares the data for prediction using machine learning techniques.

5. Visualization of the results: The predicted AQI value is displayed through a user interface. The interface presents the results in a simple and understandable format so that users can easily interpret the air quality conditions.

6. User-Friendly Interface: The system provides an easy-to-use interface where users can input data, run simulations, and view the predicted air quality results efficiently.

2.5 Other Nonfunctional Requirements

Non-functional requirements describe the quality attributes and performance expectations of the Air Quality Prediction System.

1. Performance: The system should process input data and generate AQI predictions within a short time to ensure smooth user interaction. The prediction process should be efficient and responsive.

2. Usability: The system should provide a simple and user-friendly interface that allows users to easily input pollution parameters and view the predicted AQI results without requiring technical expertise.

3. Reliability: The system should operate consistently and produce stable predictions based on the given input data. It should handle different input values without crashing or producing errors.

4. Scalability: The system should be designed in such a way that additional features, sensors, or data sources can be integrated in the future without major changes to the existing system.

5. Maintainability: The software should be structured in a modular way so that updates, improvements, or bug fixes can be implemented easily without affecting the entire system.

6. Portability: The system should be able to run on commonly used operating systems such as Windows and Linux with minimal configuration changes.

Chapter 3

System Architecture and Design

3.1 System Overview

The following figure provides a system overview of the Air Quality Prediction System, showing the overall workflow and interaction between its major components.

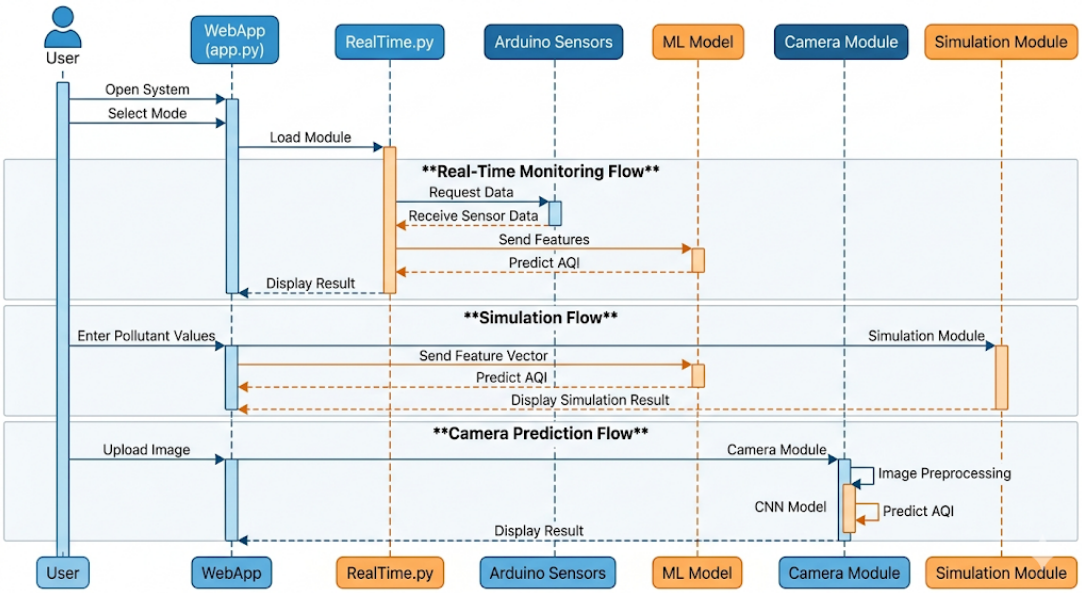


Figure 3.1: Sequence Diagram

The Air Quality Prediction System is designed to estimate air pollution levels using environmental parameters and machine learning techniques. The system analyzes pollution-related data and predicts the Air Quality Index (AQI), which represents the overall quality of air in a given environment. AQI is an important indicator that helps determine whether the air is safe or harmful for human health. The proposed system integrates multiple components such as data input, simulation, camera input, machine learning prediction, and result visualization to provide a simple and effective air quality monitoring solution.

The system collects input data in two main ways. The first method is through a simulation module, which allows users to manually enter pollution parameters such as PM2.5, CO, and NH3. This feature is particularly useful when real sensors are not available, as it allows the system to simulate pollution data for testing and demonstration purposes. By changing the values of these parameters, users can observe how different pollution conditions influence the predicted AQI. The second method of input is through a camera module, which captures environmental images. These images help provide a visual understanding of atmospheric conditions and support the environmental monitoring process.

Once the input data is collected, it is processed and prepared for analysis. The pollution parameters entered by the user or generated by the simulation module are passed to the machine learning prediction module. In this module, a trained machine learning model analyzes the input features and predicts the AQI value. The model has been trained using pollution-related data so that it can recognize patterns between pollutant concentrations and air quality levels. Machine learning enables the system to produce predictions quickly and efficiently based on the given environmental parameters.

After the prediction is generated, the results are displayed through a visual interface or dashboard. The dashboard presents the predicted AQI value in a clear and easy-to-understand format. Users can view the predicted air quality level and observe how changes in pollution parameters affect the AQI. This visualization helps users better understand environmental conditions and the impact of different pollutants on air quality.

The system architecture is designed in a modular manner, allowing different components to interact smoothly with each other. The main modules of the system include the input module, simulation module, camera module, machine learning prediction module, and visualization interface. The input module collects pollution data from the user, while the simulation module generates data for testing purposes. The camera module captures environmental images, and the machine learning module processes the input data to predict the AQI. Finally, the visualization interface displays the results to the user.

The overall workflow of the system begins with collecting environmental data through user input or simulation. The collected data is then processed and passed to the machine learning model for prediction. The model analyzes the pollution parameters and generates the AQI value. The predicted AQI is then presented to the user through the interface,

allowing users to monitor and understand air quality conditions effectively.

The proposed Air Quality Prediction System demonstrates how modern technologies such as machine learning, data processing, and visualization tools can be combined to create an intelligent environmental monitoring solution. Although the current system uses simulated pollution data, camera input and real-time sensors for demonstration purposes, it can be extended in the future to integrate advanced prediction models, and larger environmental datasets. This system can serve as a foundation for developing more advanced air quality monitoring platforms that support environmental awareness and decision-making.

The system uses pollutant concentration data such as PM2.5, CO, and NH3 along with a camera monitoring module to observe environmental conditions. These components work together to analyze pollution levels and predict the Air Quality Index (AQI).

The main components used in the system are described in Table 3.1.

Table 3.1: System Components Used for Air Quality Monitoring

Component	**Description**
PM2.5	Fine particulate matter with diameter less than 2.5 micrometers which significantly affects air quality and human health.
CO	Carbon Monoxide concentration produced mainly from vehicle emissions and industrial activities.
NH3	Ammonia gas released from agricultural and industrial processes that contributes to air pollution.
Camera Module	Captures environmental images or video frames to observe atmospheric conditions such as smoke, haze, and traffic pollution sources.
AQI	Air Quality Index representing the overall air pollution level calculated from pollutant concentration values.

Chapter 4

System Architecture and Design

4.1 Architectural Design

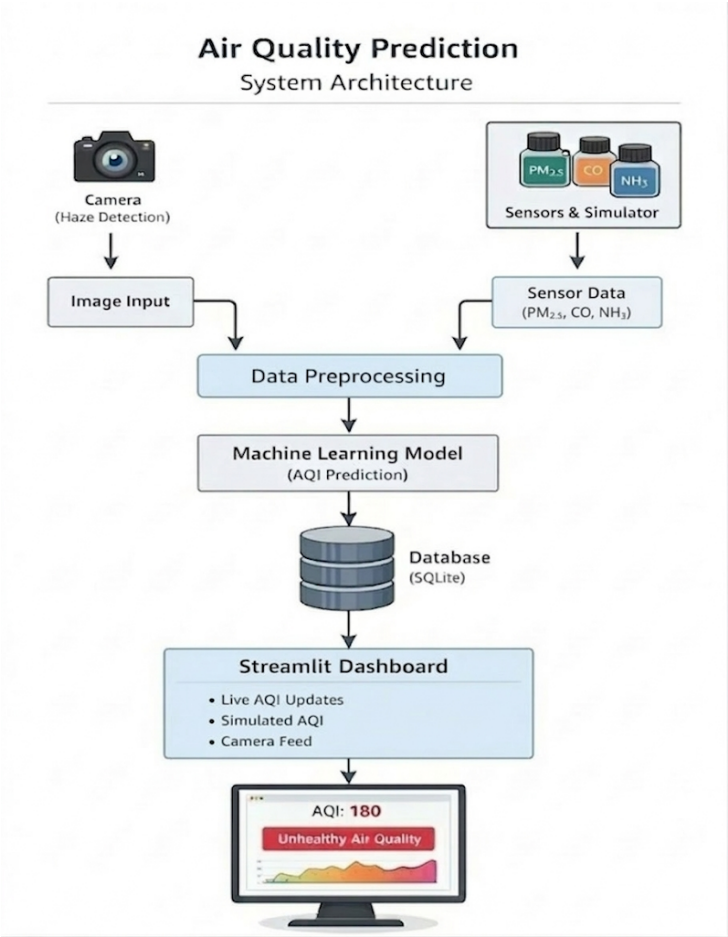


Figure 4.1: Architecture Diagram

4.2 Dataset identified

This section describes the data sources used in the development of the Air Quality Prediction system. The project utilizes both environmental image data and numerical Air Quality Index (AQI) datasets to train and evaluate the model. These datasets help the system understand environmental conditions and learn the relationship between pollution levels and visual indicators.

The environmental image dataset consists of images captured under different atmospheric conditions such as clear air, moderate haze, and heavy haze. These images help the system analyze visual patterns related to pollution using image processing and computer vision techniques. Before training the model, the images are preprocessed using techniques such as resizing, normalization, and noise reduction to improve data consistency and accuracy.

In addition to image data, the system uses an AQI dataset that contains pollutant concentration values such as PM2.5, NH3, and CO. These pollutants are important indicators of air quality and are widely used in pollution monitoring systems. PM2.5 represents fine particulate matter present in the air, NH3 (ammonia) indicates the presence of agricultural and industrial emissions, and CO (carbon monoxide) reflects incomplete combustion from vehicles and industrial activities.

Sample subsets of the dataset include environmental conditions with low pollutant levels, moderate pollution levels, and high pollution levels. The dataset is divided into training and testing sets so that the machine learning model can learn patterns from the training data and then be evaluated using unseen test data.

Table 4.1: Sample Air Quality Dataset Used for Training with Camera Observation

PM2.5 ($\mu g/m^3$)	**CO (ppm)**	**NH3** ($\mu g/m^3$)	**Camera Observation**	**AQI**
120	12.5	45	Heavy Smoke Detected	180
95	10.2	38	Moderate Haze	150
60	7.5	30	Light Pollution Visible	110
40	5.1	20	Clear Environment	80

4.3 Proposed Methodology/Algorithms

4.3.1 Methodology

This section describes the methodologies and algorithms used in the Air Quality Prediction system. The system combines image processing, machine learning techniques, and pollution dataset analysis to estimate air quality levels. The main goal is to analyze environmental images captured through a camera and correlate them with pollution data such as PM2.5, NH3, and CO to predict the Air Quality Index (AQI).

The system works in several stages including data collection, preprocessing, feature extraction, model training, and prediction. Environmental images are first captured using a camera module and processed using computer vision techniques. At the same time, numerical pollution datasets containing pollutant concentration values are used to train the machine learning model. By learning patterns from both visual environmental features and pollutant data, the system predicts the air quality condition.

Image preprocessing techniques such as resizing, normalization, and noise reduction are applied to improve the quality of the input images. These processed images are then analyzed to extract visual features that represent environmental visibility and haze conditions. These features are then passed into the machine learning model.

The machine learning algorithm is trained using pollutant datasets that include PM2.5, NH3, and CO values, which are key indicators of air pollution. After training, the model is able to predict air quality levels based on the input data. The predicted results can then be displayed through the system interface to represent the current environmental air quality condition

4.3.2 Algorithm

Air Quality Prediction System

Input: Camera image, Pollution dataset (PM2.5, NH3, CO)

Output: Predicted Air Quality Level (AQI)

Step 1: Start

Step 2: Capture environmental image using the camera module.

Step 3:Perform image preprocessing

Step 3.1: Resize the image

Step 3.2: Normalize pixel values

Step 3.3: Remove noise

Step 4: Extract visual features from the image using image processing techniques.

Step 5: Load the pollution dataset containing PM2.5, NH3, and CO values.

Step 6: Train the machine learning model using the dataset.

Step 7: Combine extracted image features with trained model inputs.

Step 8: Predict the Air Quality Index (AQI) using the trained model.

Step 9: Display the predicted air quality level in the system interface.

Step 10: Repeat the process for real-time monitoring

Step 11: Stop

4.4 User Interface Design

The user interface (UI) of the Air Quality Prediction system is designed to provide a simple and interactive platform for users to monitor and analyze air quality conditions. The interface allows users to input pollution parameters, view camera-based environmental data, and observe the predicted Air Quality Index (AQI). The design focuses on clarity, ease of use, and proper visualization of system outputs so that users can easily understand the environmental conditions.

The UI is divided into multiple sections that guide the user through different functionalities of the system.

4.4.1 System Dashboard Interface

The System Dashboard Interface provides an overview of the system. It displays the system title, navigation options, and a summary of the current air quality prediction results. The dashboard acts as the main page from which users can access other system features.

System Dashboard Interface

Figure 4.2: System Dashboard Interface

4.4.2 Camera Monitoring Interface

The Camera Monitoring Interface shows the real-time camera feed used to capture environmental images.

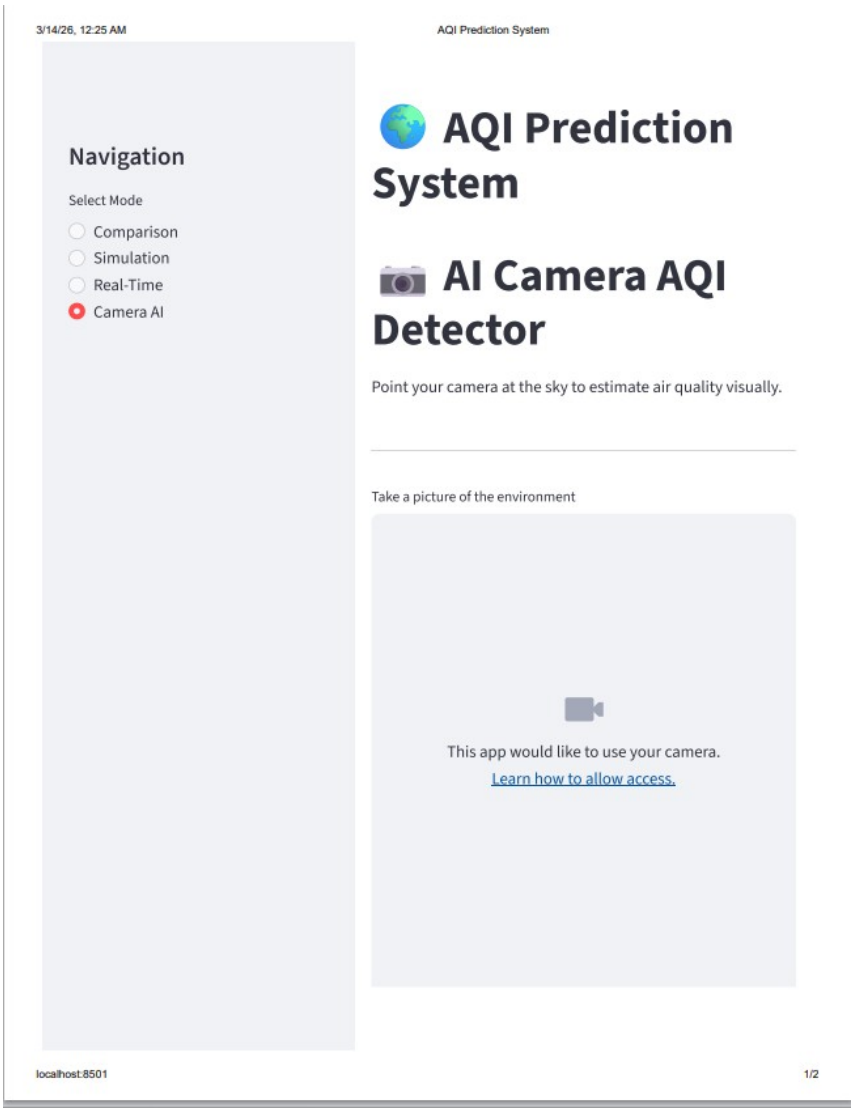


Figure 4.3: Camera Monitoring Interface

4.4.3 Pollution Simulation Interface

This interface allows users to manually enter pollution parameters such as PM2.5, NH3, and CO.

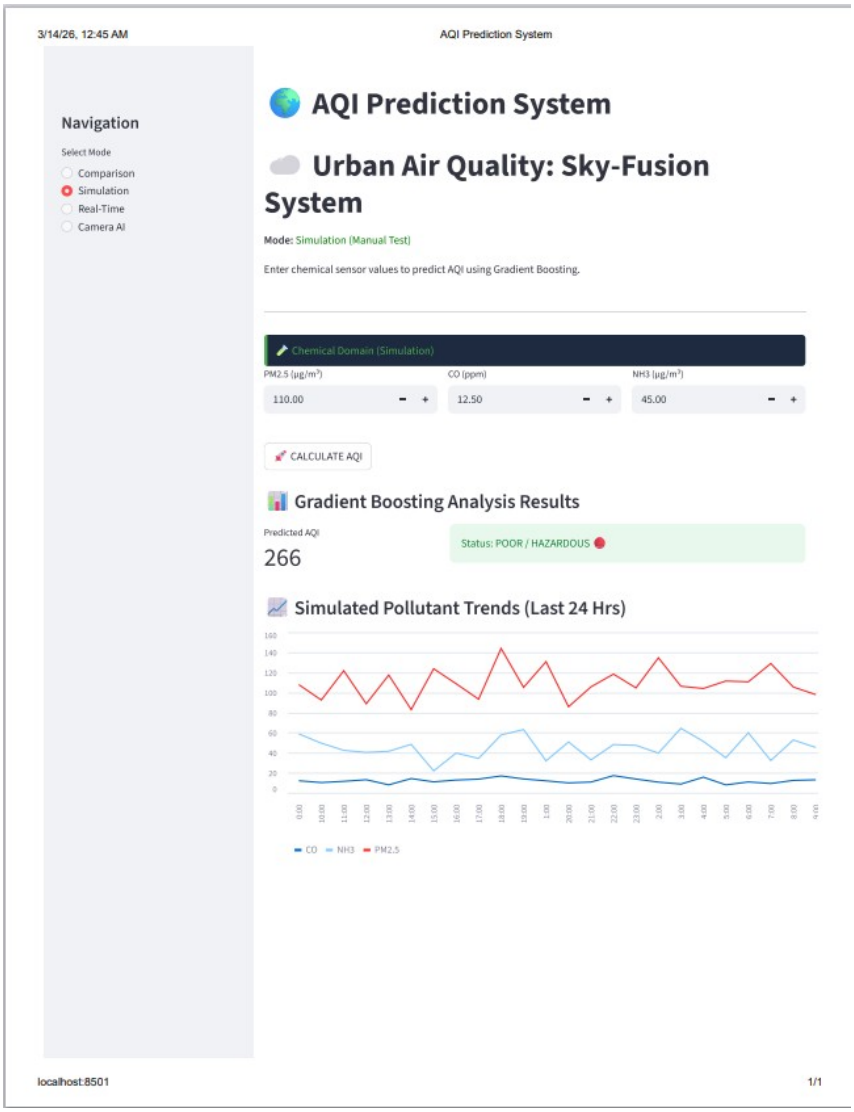


Figure 4.4: Pollution Simulation Interface

4.4.4 AQI Prediction Result Interface

The AQI Prediction Result Interface displays the predicted Air Quality Index.

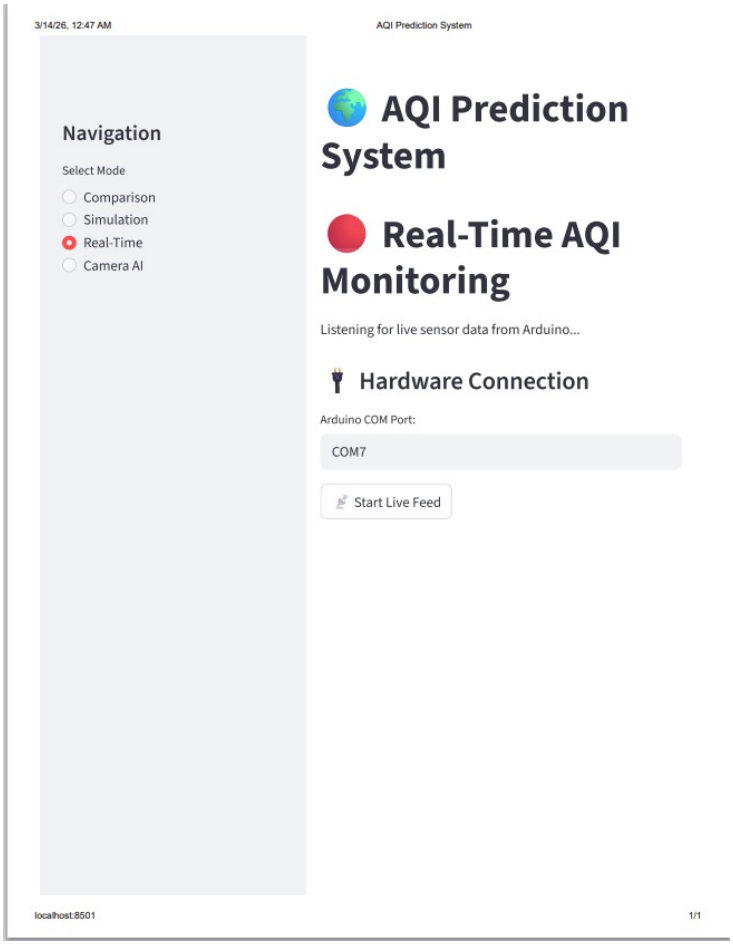


Figure 4.5: AQI Prediction Result Interface

4.5 Database Design

The Air Quality Prediction system uses a lightweight database to store environmental data, prediction results, and system logs. The database helps in maintaining historical records of pollution levels and predicted Air Quality Index (AQI) values, which can later be used for analysis and visualization. In this project, SQLite is used as the database system because it is simple to implement, lightweight, and does not require a separate server setup. It integrates easily with Python applications and is well suited for small to medium-scale projects.

The database stores pollution-related parameters such as PM2.5, NH3, CO, along with the predicted AQI value, timestamp, and the image captured by the camera module if required. These stored records allow the system to maintain a history of environmental conditions and predictions. The database also supports efficient retrieval of stored data for displaying historical trends and graphs in the user interface.

The main reason for choosing SQLite is its simplicity, portability, and minimal configuration requirements. Since the project is designed as a prototype system, SQLite provides sufficient functionality without requiring a dedicated database server such as MySQL or PostgreSQL. Additionally, SQLite works efficiently with Python libraries, making it convenient for integrating with the machine learning and image processing modules used in the system

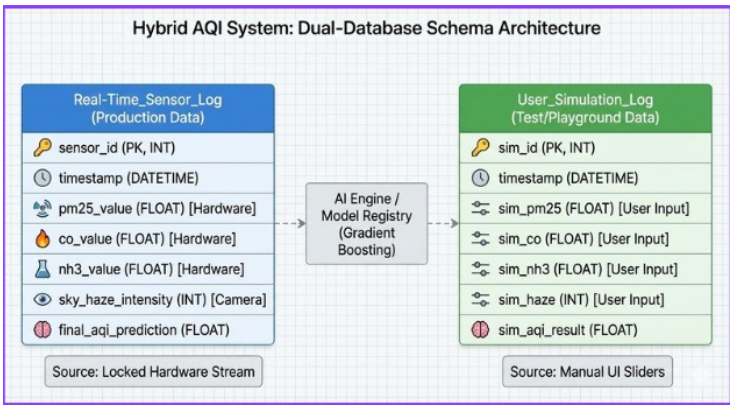
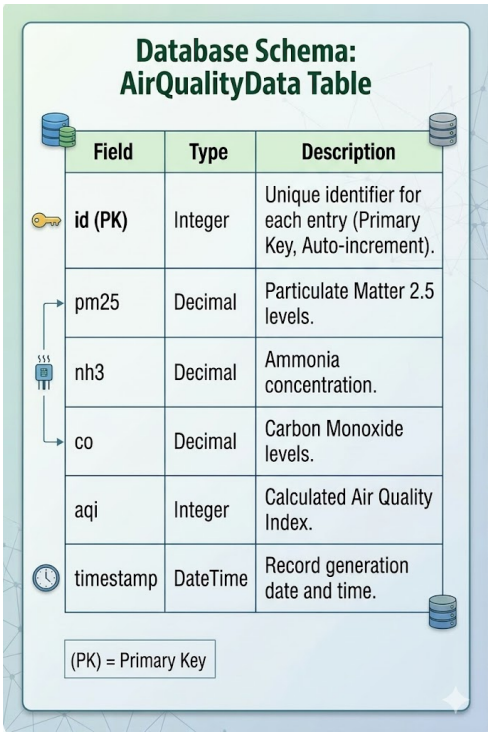


Figure 4.6: Database Schema

Figure 4.7: Database Table



Database Schema: AirQualityData Table

Field	Type	Description
id (PK)	Integer	Unique identifier for each entry (Primary Key, Auto-increment).
pm25	Decimal	Particulate Matter 2.5 levels.
nh3	Decimal	Ammonia concentration.
co	Decimal	Carbon Monoxide levels.
aqi	Integer	Calculated Air Quality Index.
timestamp	DateTime	Record generation date and time.

(PK) = Primary Key

4.6 Description of Implementation Strategies

4.6.1 Data Loading from AQI Dataset

The system uses a dataset named `Air_quality_data.csv` which contains pollutant concentration values and corresponding AQI values. This dataset is used for training and validating the machine learning model.

The Pandas library is used to load and process the dataset.

```
import pandas as pd
data = pd.read_csv("Air_quality_data.csv")
features = data[["pm2_5", "nh3", "co"]]
labels = data["aqi"]
```

4.6.2 Machine Learning Model Loading

The trained model used in the system is stored in a file named `aqi_model.pkl`. The model is loaded using the Joblib library.

```
import joblib
model = joblib.load("aqi_model.pkl")
```

4.6.3 Simulation Module

The Simulation module allows users to manually input pollutant values such as PM2.5, NH3, and CO. These values simulate environmental conditions and are used as input for AQI prediction.

```
pm25 = float(input("Enter PM2.5 value: "))
nh3 = float(input("Enter NH3 value: "))
co = float(input("Enter CO value: "))
```

4.6.4 AQI Prediction

The system predicts AQI using the trained machine learning model.

```
prediction = model.predict([[pm25, nh3, co]])
print("Predicted AQI:", prediction[0])
```

4.6.5 AQI Comparison Module

This module compares predicted AQI values with actual AQI values from the dataset.

```
actual_aqi = data["aqi"]
predicted_aqi = model.predict(features)
```

4.6.6 Model Evaluation

Model performance is evaluated using Mean Squared Error (MSE).

```
from sklearn.metrics import mean_squared_error
mse = mean_squared_error(actual_aqi, predicted_aqi)
print("Mean Squared Error:", mse)
```

4.7 Module Division

The Air Quality Prediction system is divided into several modules to ensure that each component of the system performs a specific task efficiently. Modular design helps in improving the maintainability and scalability of the system. Each module is responsible for a specific function such as data handling, prediction, simulation, and result comparison.

4.7.1 System Modules

The major modules used in this project are described below.

1. **Data Handling Module**

   This module is responsible for loading and processing the AQI dataset used in the project. The dataset contains pollutant concentration values such as PM2.5, NH3, and CO along with their corresponding AQI values. The dataset is read using Python libraries such as *Pandas*, which allows efficient data manipulation and preparation for prediction.

2. **Machine Learning Prediction Module**

   The machine learning module is responsible for predicting the Air Quality Index based on the pollutant values provided as input. A trained model stored in the dataset is loaded using the *Joblib* library. The model processes the input pollutant values and predicts the corresponding AQI value.

3. **Simulation Module**

   The simulation module allows users to manually enter pollution parameters such as PM2.5, NH3, and CO. These values simulate real environmental conditions and are used as input for the prediction model. This module helps demonstrate how different pollution levels affect the predicted AQI.

4. **Camera Monitoring Module**

   The camera module is responsible for capturing visual data from the surrounding environment using a camera device. The captured images or video frames can be used to observe environmental conditions such as smoke, haze, or traffic pollution sources. This module enhances the monitoring capability of the system by providing visual confirmation of air pollution conditions. The captured data can also be integrated with the system for future extensions involving computer vision-based pollution detection.

5. **Comparison Module**

   The comparison module evaluates the predicted AQI values against the actual AQI values available in the dataset. This helps assess the accuracy and reliability of the machine learning model used in the system.

6. **User Interface Module**

   The user interface module provides a platform for users to interact with the system. It allows users to input pollution parameters, view prediction results, and observe air quality conditions in a simple and user-friendly manner.

4.8 Work Schedule - Gantt Chart

Figure 4.8: Work Schedule - Gantt Char

Table 4.2: Machine Learning Model Performance Evaluation

Performance Metric	**Value**
Mean Squared Error (MSE)	12.4
Root Mean Squared Error (RMSE)	3.52
R-Squared Score (R^2)	0.89
Model Used	Gradient Boosting Regressor

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Appendix C: CO-PO-PSO Mapping

COURSE OUTCOMES (CO)

SNO	Description	Bloom's Taxonomy Level
1	Model and solve real world problems by applying knowledge across domains.	Apply
2	Develop products, processes or technologies for sustainable and socially relevant applications.	Apply
3	Function effectively as an individual and as a leader in diverse teams and to comprehend and execute designated tasks.	Apply
4	Plan and execute tasks utilizing available resources within timelines, following ethical and professional norms.	Apply
5	Identify technology/research gaps and propose innovative/creative solutions.	Analyze
6	Organize and communicate technical and scientific findings effectively in written and oral forms.	Apply

MAPPING COURSE OUTCOMES (COs) – PROGRAM OUTCOMES (POs) AND COURSE OUTCOMES (COs) – PROGRAM SPECIFIC OUTCOMES (PSOs)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	2	2	2	1		2	1	1	1	1	1	2	2	2
CO2	2	2	2		2	3	2	1	1	1	1	2	2	
CO3								3	2	2	1			3
CO4		1	2		3	2	3	2	2	3	1		2	2
CO5	2	3	3	1	2					1	3	3		
CO6			2			2	2	2	3	1	1			3

JUSTIFICATIONS FOR CO-PO MAPPING

Mapping	Level	Justification
101003/ CS822U.CO1- PO1	Medium	Applies engineering knowledge, mathematics, and computing fundamentals to model and solve real-world problems.
101003/ CS822U.CO1- PO2	Medium	Requires identification, formulation and analysis of engineering problems before developing solutions.
101003/ CS822U.CO1- PO3	Medium	Involves designing systems or solutions to address identified engineering needs.
101003/ CS822U.CO1- PO4	Low	Limited investigation or experimentation may be required to validate the developed models or solutions.
101003/ CS822U.CO1- PO6	Medium	Solutions developed may influence societal, environmental, or sustainability aspects of engineering applications.
101003/ CS822U.CO1- PO7	Medium	Encourages consideration of ethical and professional responsibilities while developing solutions.
101003/ CS822U.CO1- PO8	Low	Some level of teamwork may be required while working on modeling or project tasks.
101003/ CS822U.CO1- PO9	Low	Requires communication of results and findings through reports or presentations.
101003/ CS822U.CO1- PO10	Low	Involves basic planning and execution of tasks while implementing solutions.
101003/ CS822U.CO1- PO11	Low	Encourages independent learning and adapting new knowledge to solve real-world problems.
101003/ CS822U.CO1- PSO1	Medium	Uses core computer science principles to analyze problems and design appropriate technical solutions.

Mapping	Level	Justification
101003/ CS822U.CO1- PSO2	Medium	Involves algorithm design and application of software development practices to implement solutions.
101003/ CS822U.CO1- PSO3	Medium	Applies computer science fundamentals in research or innovative problem solving for societal needs.
101003/ CS822U.CO2- PO1	Medium	Utilizes engineering knowledge to design and develop products or systems.
101003/ CS822U.CO2- PO2	Medium	Requires analysis of user needs and technological feasibility before development.
101003/ CS822U.CO2- PO3	Medium	Focuses on design and development of solutions addressing real-world requirements.
101003/ CS822U.CO2- PO5	Medium	Requires use of modern engineering and IT tools for system or product development.
101003/ CS822U.CO2- PO6	High	Emphasizes sustainability and societal impact during solution development.
101003/ CS822U.CO2- PO7	Medium	Ethical and professional responsibilities are considered when developing socially relevant technologies.
101003/ CS822U.CO2- PO8	Low	Development activities may require collaboration among team members.
101003/ CS822U.CO2- PO9	Low	Involves communicating design ideas and development outcomes.
101003/ CS822U.CO2- PO10	Low	Requires basic planning and coordination during development tasks.
101003/ CS822U.CO2- PO11	Low	Encourages continuous learning of emerging technologies while building solutions.

Mapping	Level	Justification
101003/ CS822U.CO2- PSO1	Medium	Applies computer science fundamentals to design technically feasible solutions.
101003/ CS822U.CO2- PSO2	Medium	Uses programming and software development practices to implement solutions.
101003/ CS822U.CO3- PO8	High	Strongly relates to functioning effectively as a member or leader in multidisciplinary teams.
101003/ CS822U.CO3- PO9	Medium	Requires effective communication among team members during collaboration.
101003/ CS822U.CO3- PO10	Medium	Team leadership involves project coordination and task management.
101003/ CS822U.CO3- PO11	Low	Team interaction supports learning from peers and adapting new approaches.
101003/ CS822U.CO3- PSO3	High	Professional collaboration and teamwork support innovative computer science solutions.
101003/ CS822U.CO4- PO2	Low	Requires analysis and planning before executing project tasks.
101003/ CS822U.CO4- PO3	Medium	Involves designing and implementing project solutions based on planned activities.
101003/ CS822U.CO4- PO5	High	Requires effective use of modern tools and resources during project execution.
101003/ CS822U.CO4- PO6	Medium	Considers societal and environmental impact while implementing solutions.
101003/ CS822U.CO4- PO7	High	Emphasizes adherence to ethical practices and professional responsibilities.
101003/ CS822U.CO4- PO8	Medium	Task execution may involve teamwork and coordination among team members.

Mapping	Level	Justification
101003/ CS822U.CO4- PO9	Medium	Requires communication of progress, outcomes and documentation.
101003/ CS822U.CO4- PO10	High	Focuses on project management including scheduling, resource allocation and task execution.
101003/ CS822U.CO4- PO11	Low	Encourages adaptability and continuous learning during project implementation.
101003/ CS822U.CO4- PSO2	Medium	Uses software development practices to execute and manage project tasks.
101003/ CS822U.CO4- PSO3	Medium	Applies computer science knowledge in practical project development environments.
101003/ CS822U.CO5- PO1	Medium	Uses engineering knowledge to understand technological limitations and opportunities.
101003/ CS822U.CO5- PO2	High	Requires deep problem analysis and identification of research gaps.
101003/ CS822U.CO5- PO3	High	Proposes innovative design solutions to address identified gaps.
101003/ CS822U.CO5- PO4	Low	May involve limited investigation or validation of proposed ideas.
101003/ CS822U.CO5- PO5	Medium	Uses appropriate tools and techniques to analyze and develop innovative solutions.
101003/ CS822U.CO5- PO10	Low	Requires minimal planning while proposing new solution approaches.
101003/ CS822U.CO5- PO11	High	Strongly promotes independent learning and exploration of emerging technologies.
101003/ CS822U.CO5- PSO1	High	Applies computer science fundamentals to identify and solve complex technical problems.

Mapping	**Level**	**Justification**
101003/ CS822U.CO6- PO3	Medium	Requires presentation of design concepts and developed solutions.
101003/ CS822U.CO6- PO6	Medium	Communication may include discussion of societal or environmental implications.
101003/ CS822U.CO6- PO7	Medium	Requires responsible and ethical communication of technical results.
101003/ CS822U.CO6- PO8	Medium	Communication supports collaboration within team environments.
101003/ CS822U.CO6- PO9	High	Strongly relates to effective written and oral communication of technical findings.
101003/ CS822U.CO6- PO10	Low	Requires structured reporting and documentation in project management contexts.
101003/ CS822U.CO6- PO11	Low	Encourages continuous improvement in communication and presentation skills.
101003/ CS822U.CO6- PSO3	High	Communication of innovative computer science solutions and research outcomes.